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# Qualifying ten DAPHNE boards at PAB

One board at a time, with four-link streaming, Iceberg timing, power measurements, and self-trigger data

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**DUNE**  
DEEP UNDERGROUND  
NEUTRINO EXPERIMENT

 **Fermilab**

# Ten boards need one stable test, not ten simultaneous slots

## THE PROPOSAL

# Move the boards to PAB and qualify them sequentially on Iceberg.

We install one DAPHNE, run all four of its data links together, close its evidence, and only then move to the next serial number.

### **10** Boards in the campaign

Every board follows the same frozen run card.

### **01** Active board

Power, timing, configuration, and DAQ focus on one device.

### **04** Simultaneous data links

The active board must sustain full streaming on all four links.

# PAB gives us the timing and DAQ context the bench cannot

## WHY ICEBERG

### Use the infrastructure that already exists

Iceberg has the hardware needed to exercise the timing master and DUNE DAQ data-taking path together. That lets us test board behavior in the environment we actually care about.

We still keep a standalone clock routine so a board can be brought up when timing is unavailable.

## WHAT MOVES

### A ten-board campaign

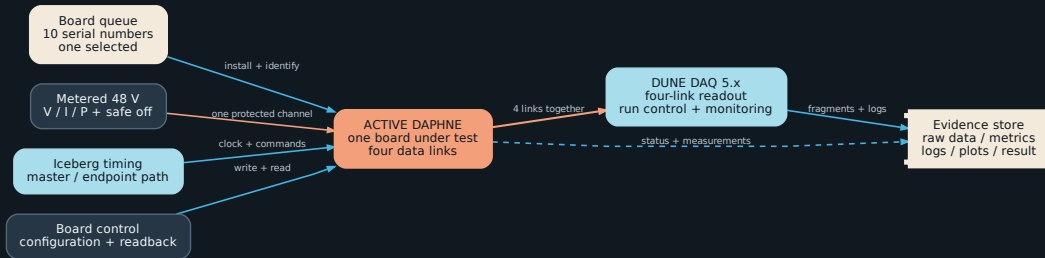
Boards, approved firmware artifacts, configuration templates, cable map, and the test guide.

## WHAT MUST BE READY FIRST

### A rehearsed DAQ 5.x workarea

One-board/four-link topology, timing commands, monitoring, data check, storage, and recovery.

# The fixture serves one active board from end to end



**Diagram.** Ten serial numbers move through one fixed fixture. The instantaneous requirement is one metered power path, one timing path, and four DAQ data links.

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PROVE THE BASELINE FIRST

# Qualify full-stream behavior before changing firmware

A known board proves the fixture; each campaign board then repeats the same configuration, timing, data, and power checks.

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# The campaign advances only after four gates pass

**01**

## **Stand ready**

Pin DAQ 5.x, map four links, check timing, power, cooling, and storage.

**02**

## **Known board**

Run the complete procedure and fix the fixture before the campaign.

**03**

## **Full stream**

Four links, timing behavior, data quality, power profile, and recovery pass.

**04**

## **Self-trigger**

Firmware update, rollback, reconfiguration, timing, and triggered data pass.

### **AFTER THE DRY RUN**

## **Freeze the fixture and run card**

During boards 01–10, only serial-specific identity and approved calibration/configuration values should change.

# Every board begins with identity and complete readback

| Check               | Save in the run directory                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Identity + firmware | <b>Exact board and image</b><br>Serial, MAC/IP, fixture position, operator, image checksum, register map. |
| Services            | <b>Small probes</b><br>Ping, board service, Hermes/IPbus magic read.                                      |
| Configuration       | <b>Write, then read back</b><br>Clock, AFE, gain, offset, threshold, mask, links, timing, sensors.        |
| Safe baseline       | <b>Idle point</b><br>Links off, clocks stable, temperatures normal, metered V/I/P.                        |

## THE FIRST GATE

### Know exactly what is installed

Serial, firmware, requested configuration, and readback must agree before streaming.

# Enable all four data links in controlled steps

**00**

## **Configured idle**

Data links disabled;  
record the first stable  
power and  
temperature point.

**01**

## **One link**

Prove endpoint  
mapping, frame  
decoding, and a  
clean start/stop.

**02**

## **Two links**

Check concurrency,  
rate, link counters,  
and the power  
increase.

**04**

## **Four links**

Stream all four  
together, then hold a  
sustained run and  
repeat start/stop.

### **A FULL-STREAM PASS**

## **Data, waveforms, and operating behavior agree**

Check fragment and sequence continuity, malformed or dropped data, representative waveforms, per-channel baseline and RMS, timestamps, input power, and temperature.

# Local timing is useful, but it is not synchronization

## STANDALONE MODE

### Prove the board can operate locally

Select the local clock. Configure and stream while reading timing status.

**Claim:** local-clock board operation works.

**Not a claim:** endpoint synchronization.

## ICEBERG-CONNECTED MODE

### Prove endpoint behavior while taking data

Select the endpoint source after clock and FPGA readiness. Verify both locks, address, ready state, and timestamp validity.

**Claim:** synchronization and usable timestamps remain stable.

## THE RULE

### Report the mode with every result

A standalone success is a valid bring-up result, but it cannot be reported as an endpoint-timing pass.

# Timing must survive commands, data-taking, and recovery

## 01 Reach ready

Select the endpoint and verify locks, address, FSM state, and valid timestamp.

## 02 Issue commands

Use the timing/run-control sequence agreed by the timing and DAQ owners.

## 03 Correlate data

Match command logs and state changes to run boundaries and decoded timestamps.

## 04 Recover once

Use one approved reset or source transition and measure the return to ready.

### PASS EVIDENCE

#### Show what happened before, during, and after

Save raw timing snapshots, the command log, decoded timestamp continuity, the observed state transition, and the recovery time.

# The power profile must show the cost of streaming

| Operating state                          | What changes                                                                            | What we record                                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Boot / safe idle                         | Firmware starts; links remain disabled                                                  | V/I/P peak, settling, temperature                                      |
| Fully configured<br>One, two, four links | Clocks and AFEs are loaded and read back<br>Transceivers and traffic are added stepwise | Steady V/I/P and temperature<br>Power delta, rate, errors, temperature |
| Four links + timing                      | Sustained stream, endpoint ready, repeated start/stop                                   | Average/peak power, data integrity, timestamps                         |
| Self-trigger                             | Quiet, threshold, and controlled-rate points                                            | Power, trigger/data rate, busy/drop                                    |

## BEFORE GATE 0

### Agree the warning and stop limits

Notes cite 48 V nominal, about 0.46 A idle, and an estimated 33 W / 0.69 A full load. These are planning references, not qualification limits (EDMS 2383681).

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THEN CHANGE THE FIRMWARE

# Treat self-trigger as a second, pinned qualification phase

Close the full-stream evidence first. Load one approved image, reconfigure the complete board, and keep a tested rollback path.

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# The firmware update needs an artifact and a rollback

## 01 Close baseline

Stop DAQ and seal the full-stream run IDs before changing the board.

## 02 Verify image

Match the approved self-trigger artifact, checksum, and register map.

## 03 Program and boot

Follow the written update procedure, then read the reported identity.

## 04 Reconfigure

Write the complete configuration and compare every possible readback.

### ROLL BACK IMMEDIATELY

#### Do not work around an inconsistent board

Restore the pinned baseline if identity, register map, configuration readback, timing readiness, power, temperature, service, or DAQ behavior is wrong.

## Self-trigger data should cover quiet, threshold, and rate points

### 01

#### Quiet baseline

Enable self-trigger with no intentional stimulus. Measure spontaneous rate, power, and counters.

### 02

#### Threshold scan

Hold the stimulus fixed and test the approved threshold points. Save waveforms and RMS.

### 03

#### Controlled rates

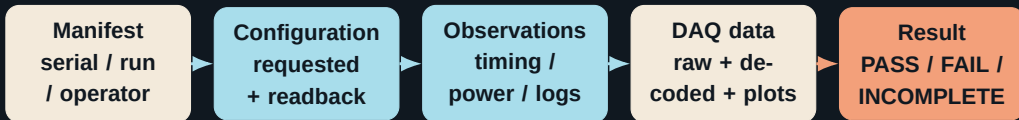
Run low, nominal, and highest approved rates; repeat one reference point at the end.

#### HOW TIMING FITS

#### Timing commands define the acquisition boundaries

Log the agreed start, stop, or synchronization commands and correlate them with self-trigger timestamps. The stimulus and threshold create photon triggers; timing commands organize the run.

# One run directory keeps every claim traceable



**Diagram.** A result is reviewable only when the serial number, firmware and configuration hashes, timing state, operating measurements, and raw data share one run ID.

## KEEP FAILED ATTEMPTS

### A successful retry gets a new run ID

Never overwrite or silently replace a failed run. Link the retest to the original failure and keep both evidence packages.

# DAQ must deliver a rehearsed 5.x workarea, not only a build

## THE DAQ ASK

## Pin one release and prove the complete operator path.

The local sources inspected for this plan contain an fddaq-v5.6.2 package baseline. The DAQ team should choose the exact 5.x tag for PAB.

### 01 Freeze compatibility

Pin daphnemodules, data formats, raw-data utilities, Hermes/readout dependencies, and both firmware maps.

### 02 Deliver the topology

Generate one-board/four-link configuration with fixed receiver, NIC, endpoint, and storage mapping.

### 03 Rehearse the operation

From a clean shell: configure, start, stop, recover, decode, report waveform/RMS, and repeat.

## The DAQ release is ready when five deliverables work together

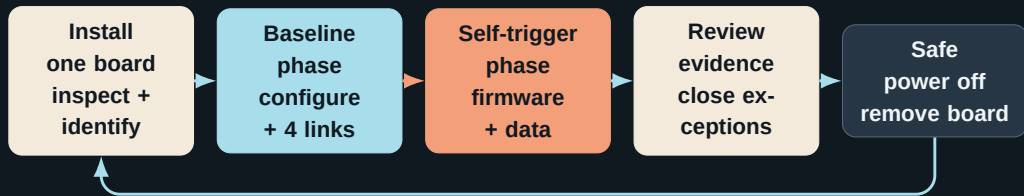
| Deliverable        | Ready means                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------|
| Release manifest   | Exact fddaq-v5.x, packages, firmware maps, and artifacts reproduce the workarea.                |
| One-board topology | Four links map to fixed receivers, interfaces, endpoints, and files without hand editing.       |
| Run-control guide  | A human can configure, start, stop, reset, and recover with short commands.                     |
| Monitoring         | Link rates/errors, fragment health, board/timing state, process and storage health are visible. |
| Data checker       | One command reports integrity, waveforms, and per-channel baseline/RMS.                         |

### THE COMPATIBILITY QUESTION

#### **Both firmware phases must be expressible**

Confirm that the chosen controller and configuration path can select the required full-stream and self-trigger modes and read back their state.

# Each serial number completes the same closed loop



**Diagram.** Board 00 qualifies the stand. Boards 01–10 then pass through sequentially; the next board is installed only after the current result is closed.

## THE CAMPAIGN RESULT

### Ten comparable board packages

Success is a consistent qualification record for every serial number. Ten-board simultaneous running is outside this plan.

# Recovery changes one thing and preserves the failure

## SERVICE OR LINK

### Check the path before the image

Verify metered power, IP/MAC, route, cable/SFP, and receiver map. Retry one small read or one link. Do not reflash first.

## POWER OR TEMPERATURE

### Stop and make the board safe

Stop streaming, save the peak and temperature, power off, and inspect cooling/configuration. No automatic re-enable.

## TIMING NOT READY

### Read the complete state

Check source, both locks, address, FSM, and timestamp validity. Reconfigure in order; local mode is diagnostic only.

## FIRMWARE OR DAQ

### Return to a pinned state

Save identity, readback, logs, and partial data. Roll back or cleanly restart; always use a new run ID.

# Approve the campaign and assign the owners

## THE DECISION

### Approve one sequential, two-phase campaign

One active board; four data links.  
Full-stream first, then pinned  
self-trigger firmware.

**Ready to move:** Board 00 passes  
both phases, recovery, decoding,  
and evidence from a clean start.

1. PAB fixture, power, cooling, and safety.
2. Timing connection and command sequence.
3. Exact DAQ 5.x release and rehearsal date.
4. Both firmware artifacts and rollback.
5. Inventory, cable map, storage, window, limits, and sign-off.

# Connected timing has a concrete readiness snapshot

| Address    | Field                      | Connected-timing acceptance                         |
|------------|----------------------------|-----------------------------------------------------|
| 0x84000000 | Clock control bit 2        | Endpoint/external timing source selected            |
| 0x84000004 | MMCM status bits 0 and 1   | Both locks asserted                                 |
| 0x84000008 | Endpoint address and reset | Expected address in the low 16 bits; reset released |
| 0x8400000C | Endpoint status            | FSM low nibble is 0x8; timestamp-valid bit 4 is set |

READ THE SNAPSHOT AS ONE CONTRACT

## Bring the path up in order

External clock, FPGA, source/reset, lock, endpoint address/reset, then ready plus valid timestamp. A common raw status is 0x18. Verify every field against the pinned firmware map.

# The baseline matrix covers idle through timed streaming

| Run | Firmware | Mode                  | Timing  | Primary evidence                                       |
|-----|----------|-----------------------|---------|--------------------------------------------------------|
| E0  | Baseline | Safe idle / configure | Local   | Identity, service, complete readback, idle V/I/P/temp. |
| F1  | Baseline | One-link full stream  | Local   | Mapping, frames, waveform/RMS, power.                  |
| F2  | Baseline | Two-link full stream  | Local   | Concurrency, errors, rate, power delta.                |
| F4  | Baseline | Four-link full stream | Local   | Four-link integrity, offered rate, V/I/P/temp.         |
| FT  | Baseline | Four-link soak        | Iceberg | Ready state, commands, timestamp continuity, recovery. |

## RUN-CARD RULE

### Every row gets a new timestamped run ID

A skipped row is INCOMPLETE with a reason. Owners still need to approve durations, offered rate, and pass/stop limits.

# The self-trigger matrix covers quiet through repeatability

| Run | Firmware     | Mode               | Timing  | Primary evidence                                |
|-----|--------------|--------------------|---------|-------------------------------------------------|
| S0  | Self-trigger | Quiet baseline     | Iceberg | Spontaneous rate, counters, power/temp.         |
| ST  | Self-trigger | Threshold points   | Iceberg | Rate response, waveforms, timestamps.           |
| SR  | Self-trigger | Controlled rates   | Iceberg | Accepted/drop/busy/full, data quality, power.   |
| SX  | Self-trigger | Repeated reference | Iceberg | Repeatability and final configuration readback. |

## RUN-CARD RULE

### Every row gets a new timestamped run ID

Timing, firmware, power, hardware, and DAQ owners must approve the stimulus, threshold/rate points, duration, and limits before Gate 0.